

Nils Duran

Machine Learning Engineer | MSc Data Science Student

📍 Barcelona, Catalonia ✉ nilsdula@gmail.com ☎ 660 05 40 68 🔗 nilsduran.github.io 🌐 nils-duran2004 🗣 nilsduran

Experience

Werfen , Software Tester Intern — Beacon Project	Barcelona, Catalonia June 2026 – present 4 months
- Engineered full-stack automated testing suites in Java for in-vitro diagnostic (IVD) laboratory software on the Beacon Project. - Implemented comprehensive API tests and FormUI end-to-end automation, verifying data consistency across clinical workflows. - Managed requirements traceability, test planning, and regulatory compliance validation using Jama Connect. - Built custom utility scripts and automation tooling that eliminated repetitive manual QA tasks and accelerated test cycles.	
Sparsity Technologies , Data Engineering Intern	Barcelona, Catalonia Fall 2025
- Engineered automated data collection and ingestion pipelines for high-performance graph database solutions. - Executed complex graph queries and data transformation workflows for competitor intelligence and marketing analytics.	

Education

MSc	Universitat Politècnica de Catalunya (UPC) , Data Science Distributed data architectures, statistical modeling, and scalable machine learning.	Barcelona, Catalonia Expected June 2028
BSc	Universitat Politècnica de Catalunya (UPC) , Artificial Intelligence Thesis on EEG cognitive decoding using Spatio-Temporal GNNs, Transformers, and LoRA.	Barcelona, Catalonia Sept 2022 – June 2026
Exchange	Karlsruhe Institute of Technology (KIT) , Computer Science Advanced distributed algorithms and high-performance systems. German C1 certified.	Karlsruhe, Germany Oct 2025 – Mar 2026

Projects

Sinergia — Multi-Agent LLMs (Telefónica Colab)	May 2025
- Designed multi-expert LangGraph agent architecture with LoRA fine-tuning, boosting MedQA accuracy by +3% absolute over Gemini Flash. - Evaluated diversity dissimilarity matrices and Elo rankings; developed interactive Streamlit app (Public Repo).	
HPC Vision Transformers on MareNostrum 5 (BSC)	June 2024
- Trained Vision Transformers on the 200 GB+ MAME dataset using the MareNostrum 5 supercomputer (top 10 public supercomputer worldwide). - Orchestrated distributed multi-GPU training with Slurm and PyTorch, achieving a 0.75 F1-score across hundreds of fine-grained categories.	
Neuroscience EEG Decoding (BSc Thesis)	June 2026
- Decoded visual consciousness states from 64-channel EEG using Riemannian geometry, Spatio-Temporal GNNs, and ST-EEGFormer with LoRA. - Integrated foundation models (LaBraM) and conducted XAI analysis via SHAP and Integrated Gradients (Public Repo).	
Vegan Tools — Production AI Web Platform	Sept 2026
- Architected and deployed full-stack platform featuring Ingredient Checker, Menu OCR, and Recipe Veganizer (Live at vegantools.org). - Engineered end-to-end multi-modal AI API integration, client state management, and custom cloud infrastructure (Public Repo).	

Skills

ML & Deep Learning Frameworks: PyTorch, Scikit-learn, LangGraph, LoRA / PEFT, Hugging Face Transformers, Gymnasium (RL), XGBoost

Engineering & Deployment: Python, Slurm (MareNostrum 5 HPC), Docker, Git/GitHub, REST APIs, Streamlit, Linux, Bash

Data & Backend: PostgreSQL, SQL, Graph Databases (Sparsity), Apache Spark, Vector Search, Automated Testing (Java)

Languages: Catalan (Native), Spanish (Bilingual), English (C2 Proficient), German (C1 Advanced)